

LESSON 13.7

REPRESENTING REAL-WORLD CONTEXTS WITH GRAPHS



LESSON INTRODUCTION

7.MA.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

LEARNING TARGETS

- I can choose an appropriate graphical representation for a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- What are the differences between numerical and categorical data?
- What graphical representations best display numerical data?
- What graphical representations best display categorical data?
- How can you determine which graphical representation to use for a given data set?

LESSON NARRATIVE

In this lesson, students determine which graphical representation is best for a given scenario/data set. This lesson begins with a review of the difference between numerical and categorical data. Next, students consider which graphical representations are designed for numerical data. They are guided through questions and fill in the blank items to help with understanding the strengths of these graphical representations and why they are used for numerical data. Students then review bar graphs and circle graphs, which are both used for categorical data. This lesson ends with a self-reflection on students' understanding of the day's lesson.

COMPONENT SUMMARY

13.7.1 WARM-UP (~5 MIN.)

Categorize data as numerical or categorical

13.7.3 BRING IT TOGETHER (3-5 MIN.)

Review graphical representations for numerical data

13.7.5 YOUR TURN! (5-7 MIN.)

Choose graphs to represent data

13.7.2 EXPLORATION (15-20 MIN.)

Choose the best graphical representation for numerical data sets

13.7.4 GUIDED INSTRUCTION (10 MIN.)

Choose the best graphical representations to represent categorical data

13.7.6 WRAP-UP (~5 MIN.)

Self-reflection on understanding of the lesson

RESOURCES

GLOSSARY

Key Terms:

- Line plot
- Histogram
- Stem-and-leaf plot
- Box plot
- Bar graph
- Circle graph

Additional Terms: N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may find it difficult to decide which graphical representation to use and may choose representations that are not best considering the type of data or size of the data set.

THINGS TO CONSIDER

- Students do not need to create graphical representations in this lesson. They are only being asked to decide which graphical representation would be best to fit each given scenario.

LITERACY AND LANGUAGE ROUTINES

Three Reads

To support reading comprehension as students work through 13.7.4 Guided Instruction, engage the class in a three-reads routine. Students must be able to understand each scenario before choosing the appropriate graphical display for the data sets. This routine supports students as they engage with the word problems and make sense of the information being presented.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

MA.K12.MTR.7.1 is embedded throughout this lesson. Data is presented through real-world contexts students will be familiar with. Encourage students to consider how they can apply what they are learning about numerical and categorical displays to other disciplines and in their everyday lives. Categorizing data is something students often do informally, so encourage them to make this connection as they work through this lesson.

DIFFERENTIATE INSTRUCTION

A reference sheet showing a visual of each type of graphical representation may be helpful to students. One side could show the graphical representations designed to be used with categorical data and give information about each one. The other side could show the four different graphical representations that are designed to be used with numerical data and give information about each of them as well. This way, students can decide which side of the sheet to use based on the type of data: categorical or numerical. Then they can make their decision from a smaller number of choices, providing an additional layer of scaffolding.

13.7.1 WARM-UP: BELL WORK

TEACHER MOVES

It is very important that students understand the difference between categorical and numerical data. In order to choose the best graphical representation for a data set, students must be able to determine if the data is categorical or numerical. Circulate as students work and review this concept, if necessary, before moving on to the rest of the lesson.



13.7.1 Warm-Up: Bell Work

To choose an appropriate graphical representation for data, the first step is to determine if the data is **numerical** (data points are numbers) or **categorical** (data points fall within categories).

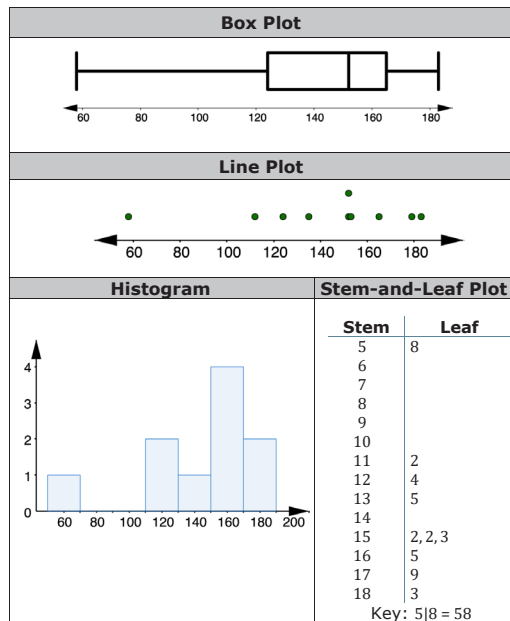
1. Complete the table. The first two rows have been completed.

Possible Question	Possible Responses	Numerical or Categorical
What is your favorite color?	red, orange, yellow, green	☒ categorical ☐ numerical
How many people are in your family?	2, 3, 4, 5...	☐ categorical ☒ numerical
How old are you?	11, 12, 13...	☐ categorical ☒ numerical
How tall are you?	5', 5'5", 5'8" ...	☐ categorical ☒ numerical
What is your favorite sport?	Football, Basketball, Baseball, Soccer	☐ categorical ☒ numerical
What is your favorite fruit?	watermelon, apple, pear, orange	☐ categorical ☒ numerical



13.7.2 Exploration: Choosing the Best Graphical Representation for Numerical Data

Four graphical representations that can be used to display **numerical data** are box plots, histograms, line plots, and stem-and-leaf plots. Each graphical representation is shown in the table.



13.7.2 EXPLORATION: CHOOSING THE BEST GRAPHICAL REPRESENTATION FOR NUMERICAL DATA

TEACHER MOVES

Notice and Wonder

Engage the class in a notice and wonder routine as they examine each graphical representation. Give students time to write their reflections, then have them share out. Record answers on chart paper or a white board. Highlight responses that reference characteristics like:

- Some graphs would be better suited for larger data sets than others
- Some data sets show spread more clearly
- Some data sets do/do not provide specific values

Work with your partner to complete the following.

- Camilla wants to create a graphical display that represents the typing speeds of the 425 students at her school.
 - Which representation(s) will best display the data?

☐ box plot
☐ line plot
☒ histogram
☐ stem-and-leaf plot
 - Explain your reasoning.
Answers will vary. Large sets of numerical data are best displayed in a box plot or histogram.
 - Camilla says, "Making a line plot of this data would take a lot of time and it would be hard to make sure each dot is in the right location on the number line, since speed can be given in fractions of seconds. I'd have to make 425 dots to represent each student!" Explain whether you agree with Camilla that a line plot would not best display the data.
Answers will vary. Camilla is correct in her explanation of why a line plot would not be the most appropriate display. It would take a lot of time to create a line plot, and because time is continuous, it would be hard to make sure every data point is accurately placed.
 - Which other display should not be considered based on the size of the data set?
Stem-and-leaf plot

TEACHER MOVES

Consider guiding students through this component to ensure they comprehend the strengths and weaknesses of each of the different graphs. As students complete each section, consider asking questions to solidify understanding:

- What are the strengths/weaknesses of [histograms, line plots, box plots, stem and leaf plots]? Why?
- When the data set is small, which graphs make sense to display the data? Why?
- When the data set is large, which graphs make sense to display the data? Why?

13.7.2 EXPLORATION: CHOOSING THE BEST GRAPHICAL REPRESENTATION FOR NUMERICAL DATA (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing sentence starters for the written explanations. This will provide an additional scaffold for students who struggle with written response. Alternatively, consider allowing students to explain their reasoning verbally to a partner.

TEACHER MOVES

Line plot and dot plot can be used interchangeably.

TEACHER MOVES

Consider providing graphs made from the data for a few of the questions. This will provide an additional scaffold for students who struggle with data visualization.

2. Dallas records the highest temperature from each month over the past year in Immokalee, Florida.

{76, 78, 81, 85, 90, 92, 92, 93, 91, 87, 82, 78}

- a. Which representation(s) will best display the data?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

- b. Explain your reasoning.

Answers will vary. A line plot is a good choice for smaller data sets. It allows every data point to be displayed, and the shape of the distribution can be determined. A box plot would provide the five-number summary of the data as well as the shape of the distribution. Stem-and-leaf may be chosen, however, it may be difficult to determine the shape of the distribution because of the small range of values.

- c. Dallas decides to change his data set to the average **weekly** high temperature. Explain whether this changes the display he should use.

Answers will vary. With 52 data points, all three above could still be used, but a box plot or a histogram would be a better choice for the large number of data points.

3. Basil recorded the weights, rounded to the nearest pound, of 15 dogs at a pet show.

7, 10, 12, 22, 28, 59, 62, 65, 72, 79, 85, 98, 100, 150, 163

- a. Explain which display would be better for the data, a line plot or a stem-and-leaf plot.

Answers will vary. Since the range of the data is large, a stem-and-leaf plot will be easier to construct since the data is displayed as an interval for each row, rather than on a line plot where the axis covers the range of data values.

- b. Basil also recorded the weights, rounded to the nearest pound, of 15 cats at the pet show.

4, 4, 5, 7, 7, 8, 8, 8, 9, 10, 11, 12, 13, 13, 14

Explain which display would be better for the data, a line plot or a histogram.

Answers will vary. A line plot would be the best choice because this is a small data set over a narrow range of values.

- c. Basil updated his data to show the weights of the cats to the nearest tenth.

4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6

Explain which display would be better for the data, a line plot or a histogram. If your recommendation is different from Part B, explain why.

Answers will vary. Because the data has been rounded to the tenth, a histogram is better. Using intervals to display the data is appropriate, since with a line plot there will be too many values on the x-axis.

4. Ms. Rallen asked her students how many pets they each have, and wants to use a graphical representation that will allow her students to see the overall shape of the data and give them enough information to calculate the 5-number summary.

- a. Which graphical representations should Ms. Rallen consider?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

- b. Explain your reasoning.

Answers will vary. The box plot, line plot, and stem-and-leaf plot will provide the information needed for a five-number summary. Specifically, the box plot provides this information best.

13.7.2 EXPLORATION: CHOOSING THE BEST GRAPHICAL REPRESENTATION FOR NUMERICAL DATA (CONTINUED)



13.7.3 Bring It Together: Choosing the Best Graphical Representation for Numerical Data



Line plots are best for smaller sets of data and when the goal is to see each value over a scale, making it easier to calculate measures of center and spread.



Histograms are best for larger sets of data, data over a large range, or continuous data. One disadvantage is that the display cannot be used to calculate measures of center and spread since individual values cannot be determined from the display.



Stem-and-leaf plots are helpful in visualizing the shape of a distribution, while still being able to see the individual values.



Box plots are helpful in visualizing the shape of a distribution and quickly identifying the five-number summary. This display cannot be used to find the mean.

13.7.3 BRING IT TOGETHER: CHOOSING THE BEST GRAPHICAL REPRESENTATION FOR NUMERICAL DATA

TEACHER MOVES

Consider adding key terms to a word wall. This will provide an additional scaffold for students and give opportunities for daily discussions.

13.7.4 GUIDED INSTRUCTION: GRAPHICAL REPRESENTATIONS FOR CATEGORICAL DATA

TEACHER MOVES

Highlight the terms in the green boxes as you work through this section with students. Encourage students to discuss scenarios in which they might want to use a bar graph or circle graph to display data. Consider asking students why these types of graphs aren't used to represent numerical data. Considering why they aren't used to represent certain types of data can help solidify why they are used for categorical data.



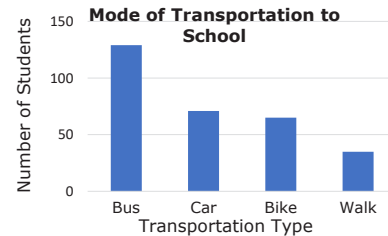
13.7.4 Guided Instruction: Graphical Representations for Categorical Data

Two graphical representations that can be used to display categorical data are bar graphs and circle graphs.

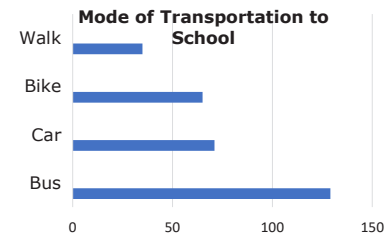


A **bar graph** is a visual display of categorical data values where each category is represented by a bar whose height represents the number in that category. Bar graphs can be represented vertically or horizontally.

The vertical bar graph shows data collected from a survey on how students get to school.



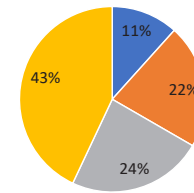
The same data is also displayed in a horizontal bar graph.



A **circle graph** is a visual display of categorical data. The whole set of data is represented by the circle and its interior. The categories are represented by fractional parts of the circle. This is also called a pie chart.

The circle graph below shows the same data.

Mode of Transportation to School



■ Walk ■ Bike ■ Car ■ Bus

1. Complete the statements.

The bar graph visually shows

- how each part relates to the whole.
- **how each part compares to the other parts.**

The circle graph shows

- **how each part relates to the whole.**
- how each part compares to the other parts.

2. Greer wants to convince the head chef at his school to include oat milk as an option at meals, so he asks a random sample of students from his school what their favorite kind of milk is: dairy, oat, or soy. He wants to use a graphical representation that makes it easy to compare the number of students who prefer oat milk to the total number of students surveyed.

Jordan asks a random sample of students from his middle school which of these apps is their favorite: TikTok, Snapchat, Instagram, Pinterest, YouTube, WhatsApp, Kik, or Discord. He wants to display the data so it is easy to compare the different categories to each other.

Complete each statement.

- ☐ Greer
☒ Jordan
- should use a bar graph. A bar graph is

the best option for this student because it

- ☐ visually shows how each part relates to the whole.
☒ visually shows how each part compares to the other parts.

- ☒ Greer
☐ Jordan
- should use a circle graph. A circle graph

is the best option for this student because it

- ☒ visually shows how each part relates to the whole.
☐ visually shows how each part compares to the other parts.

13.7.4 GUIDED INSTRUCTION: GRAPHICAL REPRESENTATIONS FOR CATEGORICAL DATA (CONTINUED)

TEACHER MOVES

Three Reads

Engage students in a Three-Reads routine as they dive into question 2. This will support reading comprehension as students make sense of what is happening in each of the scenarios presented. If students are working with a partner to complete this section, this routine also encourages students to navigate mathematical information through partner discourse. If the teacher is modeling, this can provide an opportunity to model how to use this strategy when reading a longer text.

13.7.5 YOUR TURN!

QUESTIONING

Challenge early finishers to create a stem-and-leaf plot or line plot to represent Mr. Shapiro's data. This will provide a visual to support their explanations for question 1.



13.7.5 Your Turn!

1. Mr. Shapiro's seventh grade class read more books than any other class in the middle school over the summer. The number of books read by the students in his class is shown below.

{1, 1, 3, 5, 5, 6, 6, 6, 6, 7, 8, 9, 10, 10, 11, 12, 12, 14, 14, 15, 15, 17, 17, 20, 23, 25, 27}

He wants to make a graph that shows the number of books read by each student. Which graphical representation would you recommend for this situation? Explain your reasoning.

Answers will vary. A dot plot and a stem-and-leaf plot allow for each data value to be seen and measures to be calculated. A histogram allows for the shape of the distribution to be seen over the large range of data.

2. Kendrick wants to make a graph that will show what proportion of students in his homeroom like science class best when compared to language arts, math, and social studies. Explain which graphical representation would best show this comparison.

Answers will vary. A bar graph would allow for each category to be shown individually and show how many students like each subject. From that information, proportions can be formed to provide Kendrick with the information he desires.

**13.7.6 Wrap-Up: Growth Mindset**

1. Every mistake a person makes, their brain actually grows! Take a moment to celebrate your mistakes by summarizing your favorite mistake from today.

My favorite mistake from today was ...

Answers will vary.

13.7.6 WRAP-UP: GROWTH MINDSET**DIFFERENTIATE INSTRUCTION**

Consider allowing students to verbally explain their choice or record it as an alternative means of assessment.